

Zihao Wei

5445 Stream Bank Ln, Greenbelt, MD 20770
zihaowei@umd.edu | zihaowei.com | GitHub | Google Scholar

EDUCATION

University of Maryland, College Park Ph.D. in Computer Science Advisor: Prof. Ruohan Gao	Sep. 2025 – Present
University of Michigan, Ann Arbor M.S.E. in Computer Science and Engineering Advisor: Prof. Andrew Owens	Aug. 2023 – Apr. 2025
University of Michigan, Ann Arbor B.S.E. in Computer Science and Engineering; <i>Summa Cum Laude</i>	Aug. 2021 – Apr. 2023
Shanghai Jiao Tong University B.S. in Electronic and Computer Engineering	Sep. 2019 – Aug. 2023

RESEARCH EXPERIENCE

University of Maryland, College Park <i>Research Assistant</i> Advisor: Prof. Ruohan Gao	Sep. 2025 – Present
- Developed an audio–visual language model for spoken video narration, using two-stage training to ground word-level prosody in audiovisual events and align pitch, speaking rate, and emotional expression. - Developed a physics-informed acoustic representation that integrates 3D geometry and visual appearance to synthesize object sounds at novel contact locations from images and a few impact recordings.	
Johns Hopkins University — JSALT 2026 <i>Research Assistant</i> Advisors: Profs. David Harwath and Karen Livescu	Jun. 2026 – Aug. 2026
Developed an omni encoder based on a mixture-of-experts architecture to improve multimodal understanding.	
University of Michigan, Ann Arbor <i>Research Assistant</i> Advisor: Prof. Andrew Owens	Jun. 2022 – Aug. 2025
- Designed a cluster masking strategy for vision–language pre-training that improved representation quality and training efficiency. - Developed image-conditioned masked diffusion captioning as a pre-training objective for visual representation learning. - Built a pipeline to generate and filter aligned audio–visual training data, improving multimodal representation learning.	
University of California, Santa Cruz <i>Research Assistant</i> Advisors: Profs. Cihang Xie, Yuyin Zhou and Alan Yuille	May 2022 – Oct. 2023
- Analyzed and optimized masked autoencoder training recipes to improve pre-training efficiency. - Developed a diffusion method guided by implicit neural representations to reconstruct 3D microscopy volumes from limited 2D projections. - Designed a pre-training method that exploits rotation invariance for 3D representation learning.	
Shanghai Jiao Tong University <i>Research Assistant</i> Advisor: Prof. Chengbin Ma	Mar. 2021 – Aug. 2021
Built a machine learning toolkit for battery lifetime prediction in battery management systems.	

PUBLICATIONS

* Equal contribution.

Objects as Audio-Visual Modal Sound Fields. Zisen Shao*, **Zihao Wei***, Derong Jin, Ruohan Gao. *ECCV 2026*.

Masked Diffusion Captioning for Visual Feature Learning. Chao Feng, **Zihao Wei**, Andrew Owens. *EMNLP 2025 Findings*.

Efficient Vision-Language Pre-training by Cluster Masking. **Zihao Wei***, Zixuan Pan*, Andrew Owens. *CVPR 2024*.

MicroDiffusion: Implicit Representation-Guided Diffusion for 3D Reconstruction from Limited 2D Microscopy Projections. Mude Hui*, **Zihao Wei***, Hongru Zhu, Fei Xia, Yuyin Zhou. *CVPR 2024*.

Masked Autoencoders are Secretly Efficient Learners. **Zihao Wei**, Chen Wei, Jieru Mei, Yutong Bai, Zeyu Wang, Xianhang Li, Hongru Zhu, Huiyu Wang, Alan Yuille, Yuyin Zhou, Cihang Xie. *CVPR Workshops 2024*; **Oral**.

SwinMM: Masked Multi-view with Swin Transformers for 3D Medical Image Segmentation. Yiqing Wang*, Zihan Li*, Jieru Mei*, **Zihao Wei***, Li Liu, Chen Wang, Shengtian Sang, Alan L. Yuille, Cihang Xie, Yuyin Zhou. *MICCAI 2023*; **Oral**.

A-ESRGAN: Training Real-World Blind Super-Resolution with Attention U-Net Discriminators. Zihao Wei, Yidong Huang, Yuang Chen, Chenhao Zheng, Jingnan Gao. *PRICAI 2023*.

Bridging Audio-Visual Semantics with Language-Guided Synthesis. Zihao Wei, Zixuan Pan, Yidong Huang, Ziqiao Ma, Ziyang Chen, Joyce Chai, Andrew Owens. *Technical report*.

Prosodic Video Narration from Streaming Audio-Visual Events . Zihao Wei, Jinson Jimmy, Kaizhi Qian, Hang Hua, Yang Zhang, Rogerio Feris, Ruohan Gao. *In submission*.

SELECTED PROJECTS

Objects as Audio-Visual Modal Sound Fields

- Combined 3D Gaussian Splatting and dense visual features to reconstruct object acoustics from multi-view images and a few impact recordings.
- Jointly optimized modal frequencies, damping, a spatially varying neural gain field, and residual noise to synthesize impact sounds at novel contact locations based on physics.
- Improved impact sound rendering over physics-based and data-driven baselines on two real-world datasets, supporting contact localization and object sound editing.

Masked Diffusion Captioning for Visual Feature Learning

- Trained an image-conditioned decoder to reconstruct caption tokens at sampled masking ratios, using masked diffusion as a visual pre-training objective.
- Applied diffusion-weighted reconstruction across token positions to supervise visual features, reducing reliance on auxiliary objectives.
- Evaluated ViT representations using linear probing and compositionality benchmarks, achieving performance competitive with autoregressive captioning and contrastive learning.

Efficient Vision-Language Pre-training by Cluster Masking

- Grouped image patches by cosine similarity of normalized RGB values and masked clusters around randomly sampled anchors, without an external grouping model.
- Reduced visual token processing while encouraging contextual inference during image–text contrastive pre-training.
- Improved representation quality over random masking, including FLIP, and evaluated zero-shot classification, linear probing, image–text retrieval, and language compositionality.

RESEARCH INTERESTS AND SKILLS

- **Multimodal Learning:** Self-supervised Learning, Multimodal Foundation Models, Multimodal Language Models
- **Generative Models:** Diffusion Models, Speech Generation, 3D Reconstruction, Spatial Audio Reconstruction
- **Tools:** Python, C++, PyTorch, Git, Docker, Google Cloud, AWS, CodeX

TEACHING AND SERVICE

Reviewer: TPAMI, CVPR, ICCV, ECCV, ICLR, NeurIPS, AAI, WACV, UIST.

Teaching: CMSC 848M: Multimodal Computer Vision (UMD, Spring 2026); CMSC 498K: Introduction to Multimodal Deep Learning (UMD, Fall 2025); EECS 376: Foundations of Computer Science (UMich, Winter 2022).

HONORS AND SCHOLARSHIP

Summa Cum Laude (2023); MCM Finalist, top 1% (2021); Dean’s List (2021–2023); University Honors (2021–2022); Academic Excellence Scholarship (2020).